

Seminars 3 and 4: Spectrum Representation

Topics: two-sided spectra, complex amplitudes, sums of sinusoids, periodicity and harmonic relations, beat notes, amplitude modulation, Fourier-series analysis and synthesis, rectangular pulse trains, Fourier-series properties, time-frequency representations, spectrograms, and chirp signals.

Key idea. A spectrum identifies the frequencies, magnitudes, and phases of the sinusoidal components of a signal. Periodic signals have harmonically related spectral lines, while spectrograms show how frequency content changes with time.

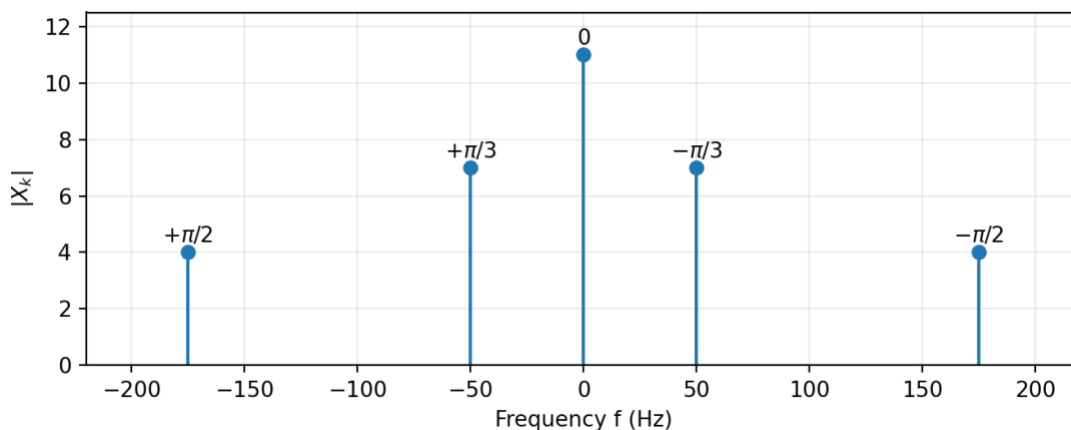
1. From a sum of sinusoids to a two-sided spectrum

Consider $x(t) = 4 + 6\cos(2\pi 100t + \pi/3) + 3\sin(2\pi 250t)$.

- Rewrite every sinusoidal term as a sum of complex exponentials.
- List all frequency–complex-amplitude pairs (f_k, X_k) , including the DC component and the negative-frequency components.
- Draw separate two-sided plots of magnitude and phase.
- Verify the conjugate-symmetry relation required for a real signal.
- Determine whether $x(t)$ is periodic. If it is, find its fundamental frequency and period.

2. From a spectrum to a time-domain signal

The following two-sided spectrum is given. The number above each line is its magnitude, and the annotation gives its phase.



Given two-sided spectrum

- Write $x(t)$ as a sum of real cosines.
- Identify the DC value.
- Determine whether $x(t)$ is real-valued and explain how the spectrum confirms this.
- Determine the fundamental frequency and fundamental period.
- Explain why the negative-frequency lines are necessary.

3. Harmonic and nonharmonic sums

Consider the two signals

$$x_1(t) = \cos(2\pi 100t) + 0.5\cos(2\pi 300t) + 0.25\cos(2\pi 500t),$$

$$x_2(t) = \cos(2\pi 100t) + 0.5\cos(2\pi 310t) + 0.25\cos(2\pi 500t).$$

- Sketch the positive-frequency magnitude spectrum of each signal.
- Determine whether each signal is periodic.
- For every periodic signal, determine the fundamental frequency and period.
- Explain the relation between harmonic spacing and periodicity.
- Plot both signals over 40 ms and compare their repetition patterns.

4. Beat notes produced by adding close frequencies

Let $x(t) = \cos(2\pi 440t) + \cos(2\pi 446t)$.

- Use a sum-to-product identity to express $x(t)$ as a high-frequency cosine multiplied by a slowly varying envelope.
- Determine the centre frequency f_c and deviation frequency f_d .
- Determine the two positive-frequency spectral components and their negative-frequency counterparts.
- Determine the perceived beat rate.
- Sketch the waveform and both envelope curves over one second.

5. Spectrum of a product of sinusoids

Consider $x(t) = 3\cos(2\pi 50t)\cos(2\pi 400t)$.

- Use a product-to-sum identity to rewrite $x(t)$ as a sum of sinusoids.
- Determine all spectral-line frequencies and complex amplitudes.
- Sketch the two-sided magnitude spectrum.
- Explain why neither 50 Hz nor 400 Hz appears as a spectral line.
- Determine whether the resulting signal is periodic and find its fundamental period.

6. Amplitude modulation

An amplitude-modulated signal is $x(t) = [2 + 0.6\cos(2\pi 120t)]\cos(2\pi 1000t)$.

- Expand $x(t)$ as a sum of three cosines.
- Identify the carrier and the two sideband frequencies.
- Determine the complex amplitude at every positive and negative frequency.
- Draw the two-sided magnitude spectrum and label the carrier and sidebands.
- State whether the envelope reaches zero and compare this signal with a beat-note waveform.

7. Finite Fourier-series synthesis

Consider $x(t) = 10 + 20\cos(2\pi 100t + \pi/4) + 10\sin(2\pi 250t)$.

- Find the greatest common fundamental frequency f_0 and the corresponding period T_0 .
- Express the signal in the finite complex Fourier-series form $x(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}$.
- Determine the nonzero coefficients a_k and their harmonic indices k .
- State a suitable value of N .
- Draw the Fourier-series spectrum versus harmonic index and versus frequency in hertz.

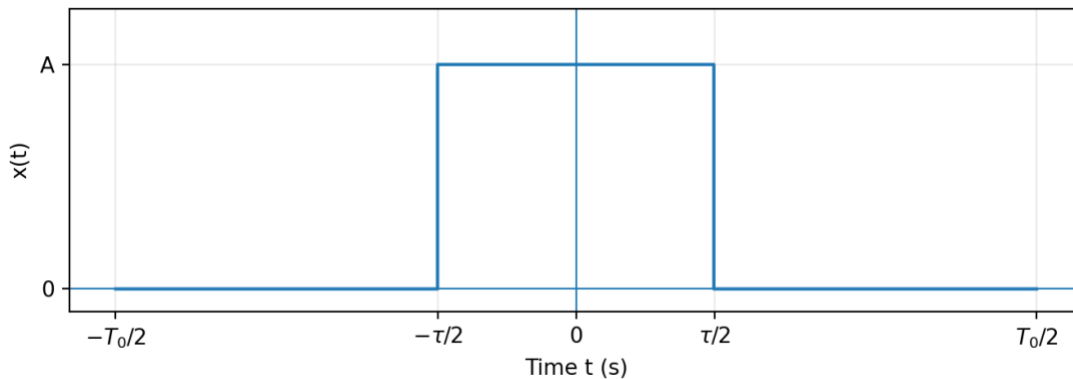
8. Fourier analysis and synthesis equations

A periodic signal has fundamental period T_0 and radian frequency $\omega_0 = 2\pi/T_0$.

- Write the complex Fourier-series synthesis equation.
- Write the corresponding Fourier-analysis equation.
- Show that a_0 is the average value of the signal over one period.
- State the conjugate-symmetry property of the coefficients for a real-valued signal.
- Explain, in words, the difference between Fourier analysis and Fourier synthesis.

9. Fourier series of a rectangular pulse train

One period of a periodic pulse train is shown below.



One period of a rectangular pulse

It is described by $x(t) = \begin{cases} A, & |t| \leq \tau/2, \\ 0, & \tau/2 < |t| \leq T_0/2, \end{cases}$

and extended periodically with period T_0 .

- Compute the DC coefficient a_0 .
- Derive a_k for $k \neq 0$ from the Fourier-analysis integral.
- Express the result using $\text{sinc}(u) = \sin(\pi u)/(\pi u)$.
- Sketch $|a_k|$ for duty cycle $\tau/T_0 = 1/2$ and $-8 \leq k \leq 8$.
- Identify which harmonics vanish and explain why.

10. Reconstructing a rectangular waveform

Use the Fourier-series coefficients from Exercise 9 with duty cycle $1/2$.

- Construct approximations using harmonic limits $N = 1$, $N = 3$, $N = 9$, and $N = 25$.
- Plot the four approximations over two periods together with the exact waveform.
- Describe how the reconstruction changes as N increases.
- Identify where the largest approximation errors occur.
- Relate the slow coefficient decay to the sharp discontinuities of the waveform.

11. Spectrum of a nonlinear sinusoidal expression

Let

$$x(t) = \cos^3(\omega_0 t).$$

- Use a trigonometric identity or complex exponentials to express $x(t)$ as a sum of cosines.
- Determine all nonzero Fourier-series coefficients a_k .
- Draw the spectrum versus harmonic index k .
- State which harmonics are absent.
- Verify that the resulting spectrum has conjugate symmetry.

12. Operations on periodic signals and Fourier coefficients

Suppose $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$.

For each operation, express the Fourier-series coefficients b_k of $y(t)$ in terms of a_k .

- $y(t) = C x(t)$, where C is a constant.
- $y(t) = x(t - t_1)$.
- $y(t) = x(-t)$.
- $y(t) = dx(t)/dt$.
- $y(t) = x(t) + x(t - T_0/2)$.

13. Reading and interpreting a spectrogram

A recording contains, in order: a constant pure tone, a rising chirp, two simultaneous harmonic tones, and a short broadband impulse.

- Describe the expected appearance of each event in a spectrogram.
- State what the horizontal and vertical axes represent.
- State what colour or intensity represents.
- Explain how a stationary spectrum differs from a time-frequency spectrum.
- Draw a labelled qualitative spectrogram for the complete recording.

14. Instantaneous frequency of a chirp

Consider the chirp $x(t) = \cos(2\pi[200t + 300t^2])$, $0 \leq t \leq 2$ s.

- Identify the time-varying phase $\Phi(t)$.
- Compute the instantaneous radian frequency $\omega_i(t) = \frac{d\Phi(t)}{dt}$.
- Compute the instantaneous frequency $f_i(t)$ in hertz.
- Determine the starting and ending frequencies.
- Plot $x(t)$ and $f_i(t)$, and sketch the corresponding spectrogram ridge.

15. Integrated spectrum-analysis exercise

Using Python, analyse $x(t) = 1.5\cos(2\pi 300t) + \cos(2\pi 306t) + [1 + 0.4\cos(2\pi 40t)]\cos(2\pi 900t)$.

- Expand the AM term and list every positive-frequency component.
- Construct the complete two-sided spectrum, including magnitudes and phases.
- Plot the waveform over 0.5 s and identify the beat behaviour.
- Produce a spectrogram and explain which components appear as horizontal lines.

- (e) Replace the 900 Hz carrier by a linear chirp from 700 Hz to 1100 Hz and describe the change in the spectrogram.